

# claude-windows / 144a31b7-7962-4fee-9d6f-8a7854daa63f

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\144a31b7-7962-4fee-9d6f-8a7854daa63f\subagents\workflows\wf_134043cd-8c3\agent-a29ce030fc623f452.jsonl`  
- SHA-256: `084dd8fa7c292071e908101db2847ae30480201968f65ef54c239049f1bc52ab`  
- Source modified: `2026-06-20T09:01:47+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_134043cd-8c3`  
- session_id: `144a31b7-7962-4fee-9d6f-8a7854daa63f`
```

## Transcript

### 1. user (2026-06-20T08:59:23.237Z)

IMPORTANT grounding (verified this session): the engine is now DECOUPLED ? pluggable StorageBackend (local + S3/GCS/DO Spaces) in backend/storage/, and a worker `python -m backend.worker {infer|train}` (backend/worker/`\_\_main\_\_`.py) that pulls from storage -> runs the pipeline -> pushes results. Validated e2e on the CPU droplet 168.144.126.176 reading/writing DO Spaces with the GPU MOCKED (stub detector, RALLY\_DETECTOR\_IMPL=stub). BOTH the khelsutra site (/srv/khelsutra-site) AND the engine (~/.rally) already live on that ONE droplet. The droplet is CPU-only (no GPU) and a SHARED testbed (MinIO 9000/9001, GCS emu 9023) ? do not propose touching those. Real GPU WASB (NativeWasbRunner, "M3.5") is the NEXT build, not done. The FastAPI has NO auth (CORS '\*'). EXCLUDE any path under .claude/worktrees/ ? those are stale copies; read canonical docs/ + backend/ only.

Map the khelsutra UI/site repo and doc conventions (cwd = C:\Users\avidu\Projects\khelsutra; the engine repo is at C:\Users\avidu\Projects\badminton-highlight-indexer). Read (in khelsutra) README.md, web/README.md, docs/PER\_RALLY\_SELECTION.md (just merged), docs/PORTABILITY.md, site/ structure; and (in engine) docs/PROJECT\_DOC\_TEMPLATE.md for the tracked-doc structure. Report: the khelsutra UI surface and how an appliance "operator console" UI would relate to the public marketing site (gated app vs public page); how this appliance relates to the per-rally selection doc (per-rally is ONE screen of the console); the exact tracked-doc template to follow; and concrete reuse opportunities. Quote path:line. Return the schema.

### 2. assistant (2026-06-20T08:59:25.667Z)

I'll map the khelsutra repo and the engine doc conventions. Let me start by pulling fresh and reading the key files in parallel.

### 3. user (2026-06-20T08:59:31.180Z)

Already up to date.

```
---BRANCH---
```

```
master
```

```
---LATEST---
```

```
eeb8a48 ci(site): add deploy-config + static-asset guards and a live-parity workflow (#12)
```

```
cb74ce5 docs(web): tracked design for interactive per-rally selection (RallyPicker) (#11)
```

```
0f555ad fix: deterministic LF line endings + live-parity check (droplet vs Cloudflare drift) (#10)
```

```
2c5159d docs: refresh NEXT_STEPS to shipped state + capture new product ideas (#9)
```

```
8591ee3 feat(site): email-on-registration via Email Routing send binding (best-effort, portable) (#8)
```

### 4. user (2026-06-20T08:59:31.203Z)

```
1      <!--
```

```
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
```

```

3
4   A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5   ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6   archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7   docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9   Rules of thumb:
10  - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11  - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12    or [design] (proposed). The doc reflects real shipped state, never aspirational.
13  - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14  - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15    research- or security/privacy-heavy work). Delete sections that do not apply.
16  - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17  -->
18
19  # <Project / Plan Title>
20
21  > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22  > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23  > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24  > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25  > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27  ---
28
29  ## 0. TL;DR
30  <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32  ## 1. Problem & goal
33  <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35  ## 2. Decisions locked
36  <Table ? capture answered questions so they are not relitigated.>
37
38  | # | Decision | Source / date | Implication |
39  |---|-----|-----|-----|
40  | D1 | ? | ? | ? |
41
42  ## 3. Foundation ? research / prior art *(OPTIONAL)*
43  <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45  ## 4. Design / architecture
46  <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48  ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49  <Who/what is protected vs NOT; residual risks stated plainly.>
50
51  ## 6. Honest limits ? what this does NOT do
52  <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>

```

```

53
54     ## 7. Deliverables & progress tracker   ? **source of truth**
55     Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
56     branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
57     | ID | Deliverable | Depends on | Gated? | Status | PR |
58     |----|-----|-----|-----|-----|----|
59     | P0 | ? | ? | No | ? | ? |
60
61     ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
62     who decides)*
63     1. ?
64
65     ## 9. Definition of done
66     <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
67     - [ ] ?
68
69     ## 10. References
70     **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
71     **External:** `<cited sources>`.
72
73     ---
74
75     ### Changelog
76     - `<YYYY-MM-DD>` ? `<what changed>`.

```

## 5. user (2026-06-20T08:59:31.212Z)

```

1     # KhelSutra Guru ?
2
3     **Private product repo** ? the heart and brain of
4     [khelsutra.guru](https://khelsutra.guru):
5     the web app (alpha SPA) now, the hosted control plane at beta.
6
7     - **Engine** (pipeline, API, eval): sibling repo `badminton-highlight-indexer` (rename
8     pending).
9     - **Canonical planning docs** (live in the engine repo's `docs/` for now, migrate here
10    at beta):
11        `UI_PROPOSAL.md` . `HOSTING_PLAN.md` . `UI_ALPHA_EXECUTION_PLAN.md`.
12
13    - **Status:** pre-alpha scaffold. SPA lands in `web/` per execution-plan milestone M1.
14
15    ## Structure
16
17    - `site/` ? public marketing landing page (static; served at khelsutra.guru)
18    - `web/` ? React SPA (alpha; Vite + React + TS + Tailwind; Cloudflare Pages build root)
19    - `docs/` ? product docs as they migrate from the engine repo
20    - `control-plane/` ? (beta) Cloud Run FastAPI control plane
21
22    ## License
23
24    All rights reserved (private; no OSS license).

```

## 6. assistant (2026-06-20T08:59:36.156Z)

Now let me read the web/ docs, the per-rally selection doc, portability, and explore the site/ and web/ structure.

## 7. user (2026-06-20T08:59:36.654Z)

```

1     SPA scaffold lands here at execution-plan milestone M1 (Vite + React + TS + Tailwind).
2     Dev: talks to the engine at http://localhost:8000 via Vite proxy; no auth locally
3     (DEV_USER_EMAIL).

```

## 8. user (2026-06-20T08:59:37.602Z)

```

1      # Interactive per-rally selection ? reel builder (RallyPicker)
2
3      > **Status:** `IN PROGRESS` ? design **APPROVED by owner 2026-06-20**; building the MVP
(P1?P5). Downstream milestones (multi-video auth, per-rally preview endpoint, any `/api/compile`
default change) remain owner-gated. . **Owner:** Avi . **Created:** 2026-06-20 . **Last
updated:** 2026-06-20
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11     ## 0. TL;DR
12     Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally
**list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest
first") pre-selects into the **same** selection set; a sticky footer compiles the selection into
one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ?
`POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396,
307-328]` ? with **zero new engine intelligence and zero new endpoints**. The single most
important caveat: queries and chips are scoped to **length / count / order / shot-count only**;
shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be
faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the
**final compiled reel** can stream ? so MVP selects on text metadata, and visual preview/timeline
is a deliberate later milestone.
13
14     ## 1. Problem & goal
15     Today the alpha can detect rallies and stitch a reel, but reel-building is either a
**silent `stitching.min_shot_count` threshold** `[verified backend/main.py:362-383]` or a static
checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies`
while the API returns `play_segments`, so it renders nothing `[verified
backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is **dynamic per-
video selection, not a fixed threshold**: a coach should say "give me the long rallies from this
session" and get exactly those, then fine-tune by hand.
16
17     **The concrete outcome ("good"): A coach uploads a 40-minute session; the engine finds
60 rallies. They type **"top 15 longest", eyeball the auto-selected cards, untick two warm-up
rallies, name it **"Tuesday ? long points", and download a ~6-minute reel ? without learning any
threshold knob.
18
19     **Goal:** the smallest **honest** surface that turns engine-detected rallies into a user-
curated, downloadable reel, with both **fast** (query) and **precise** (checkbox) selection
sharing **one** state.
20
21     **Non-goals (alpha):** no conversational LLM (`UI_PROPOSAL.md:66` defers it); no shot-
type/player/score filters (no data exists); no per-rally video preview while it requires unbuilt
streaming; no multi-tenant auth (single-box alpha).
22
23     ## 2. Decisions locked
24     | # | Decision | Source / date | Implication |
25     |---|-----|-----|-----|
26     | D1 | Dominant surface = selectable rally **list/grid** | Synthesis 2026-06-20 (Grid

```

scored 15, fit-to-data 4) | Scales to long matches via bulk+filter; shows honest fields only |  
27 | D2 | Fast entry = **\*\*deterministic\*\*** client-side `parseQuery()` (no LLM) | `[verified  
UI\_PROPOSAL.md:66]` conversational QA deferred | Predictable, unit-testable, offline, zero API  
cost |  
28 | D3 | **\*\*One\*\*** shared `Set<segment\_id>`, mutated by query + checkbox + (future) timeline  
| Synthesis 2026-06-20 | No second compile path can diverge from what the user sees |  
29 | D4 | Compile via existing `POST /api/compile {video\_id, segment\_ids, output\_filename}`  
| `[verified backend/main.py:342-396]` | Selection order is harmless ? stitcher  
`\_merge\_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]` |  
30 | D5 | Query grammar (alpha) = length, count, order, **\*\*shot\_count only\*\*** | `[verified  
database.py:122-135, gemini.py:470-485]` | Only fields the payload actually returns; grey out  
the rest |  
31 | D6 | MVP backend = **\*\*zero new endpoints\*\*** | `[verified]` reuse map (§4) | Ships the  
full loop on the already-shipped contract |  
32 | D7 | Replace the silent `min\_shot\_count` cut with **\*\*explicit user selection\*\*** | Owner  
framing + `[verified main.py:362-383]` | UI **\*warns\*** about the floor; it never hides why a rally  
vanished |  
33 | D8 | **\*\*Default selection = ON\*\*** (curate-by-removal) | Owner 2026-06-20 | Rallies load  
all-selected (matches today's "all rallies" reel); user removes warm-ups. Footer/empty UX  
assumes a non-empty start |  
34 | D9 | **\*\*Preview-before-select is NOT a ship-blocker\*\*** for the MVP | Owner 2026-06-20 |  
MVP selects on text metadata + previews the **\*final compiled reel\***; the source/clip-stream  
endpoint (per-rally preview) is the next milestone (P9), not MVP |  
35 | D10 | **\*\*MVP is job-tethered\*\*** (carry `result.video\_id`); no `GET /api/videos` | Owner  
2026-06-20 | Smallest honest slice; multi-video home + identity scoping is P8 (gated) |  
36 | D11 | `min\_shot\_count`: **\*\*warn, do NOT override\*\*** in the MVP | Owner 2026-06-20 | The  
UI surfaces the floor; an explicit-selection override of the compile default is deferred (would  
trigger the Launch Report convention) |  
37  
38 **## 3. Foundation ? what the engine gives us today** `[verified this session]`  
39 The whole design rests on the engine's real, already-shipped contract. Read against the  
engine repo (`badminton-highlight-indexer`):  
40  
41 - **\*\*The rally store is `play\_segments`\*\*** ? columns `id, video\_id, start\_time, end\_time,  
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the  
UI selects and passes to compile; `start\_time/end\_time/duration` are REAL seconds (duration  
computed at insert).  
42 - **\*\*Listing rallies = `POST /api/query {video\_id}` ? `{play\_segments:[...]}`\*\*** ? the  
**\*only\*** rally-list path today; it JSON-parses `metadata` and joins `events` `[verified  
main.py:331-340, database.py:427-475]`.  
43 - **\*\*Building a reel = `POST /api/compile {video\_id, segment\_ids:[int],  
output\_filename}`\*\*** ? loads `videos.filepath`, filters all segments to the requested  
`segment\_ids`, applies the `min\_shot\_count` floor, extracts `time\_pairs=[(start,end)]`, and  
calls `VideoCompiler.compile\_highlights(raw\_path, segments, name)` `[verified main.py:342-396,  
compiler.py:303]`, which **\*\*merges overlaps\*\*** then per-clip re-encodes ? concat ? one MP4  
`[verified compiler.py:311-317]`. Selection order/duplicates are therefore safe.  
44 - **\*\*Downloading = `GET /api/highlights/{video\_id}/{filename}`\*\*** ? streams the compiled  
MP4. Compile returns a **\*\*server-local filepath, NOT a URL\*\*** `[verified main.py:394]`, so the SPA  
constructs this download URL itself. The compiled reel is the **\*\*only streamable video that  
exists today\*\*** `[verified main.py:307-328]`.  
45 - **\*\*The floor knob is readable\*\*** ? `GET /api/config` exposes `stitching.min\_shot\_count`  
`[verified main.py:121-123]`, so the UI can warn before a selected rally is silently dropped.  
46 - **\*\*Job entry\*\*** ? `GET /api/jobs/{job\_id}` returns `result.video\_id` `[verified  
main.py:274-293]`, which is how the MVP reaches a video without a (nonexistent) list-videos  
endpoint.  
47  
48 **\*\*Honest fields vs fabricated/roadmap (critical for the UI):\*\***  
49 - **\*\*Honest today\*\*** (default `default\_segmenter: "gemini"`, per `config.json`  
`[verified]`): `start\_time`, `end\_time`, `duration`, and `metadata.shot\_count` (Gemini's  
`shuttle\_exchange\_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot\_count\_confidence` (a  
**\*\*string\*\***, often `"Unknown"`) `[verified gemini.py:470-485]`. On the gemini path  
`server/receiver` are written `None` ("no fabricated data") `[verified gemini.py:480-487]`.  
50 - **\*\*Fabricated ? do NOT surface:\*\*** the `motion` demo segmenter fills `server/receiver`  
via `random.sample(...)` , a random `shot\_count`, and random `events` (serve/smash/foul/winner)  
`[verified motion.py:191-198]`. `query` returns these columns + `events`, but they are not

ground truth and must stay hidden.

51 - **\*\*Roadmap ? no column, no detector:\*\*** shot-type per shot, player identity, match/point score, game/set grouping. None exist anywhere in the schema; do not consume them ``[verified database.py whole-schema 122-148]``.

52 - **\*\*Correction-loop columns exist but are dead:\*\*** ``candidate_segments.human_label/notes/confirmed_segment_id`` are defined but no endpoint writes them ``[verified database.py:166-168]`` ? that's the M4 corpus extension (engine PR-4/B6, ``UI_ALPHA_EXECUTION_PLAN.md``).

53

54 **## 4. Design / architecture**

55

56 **\*\*Reuse map (existing ? ``[verified]``, no change for MVP):\*\*** ``POST /api/query`` (list) ? ``POST /api/compile`` (stitch) ? ``GET /api/highlights/{video_id}/{filename}`` (download); ``GET /api/config`` (floor) ; ``GET /api/jobs/{id}`` (carry ``result.video_id``). **\*\*All net-new code is the SPA\*\*** in ``khelsutra`` ``web/`` (Vite + React + TS + Tailwind, per ``web/README.md``), plus a small set of *optional* real-gap engine enablers (§4.3) ? none required for MVP.

57

58 **\*\*4.1 Components (net-new SPA)\*\***

59 - **\*\*QueryBar + ``parseQuery()`` \*\*** ? a *pure, deterministic, unit-tested* string ? ``{filter, sort}`` compiler over ``duration`` / ``ordinal`` / ``shot_count``. Renders a **\*\*parsed-pill\*\*** (`"filter: duration > 10s · sort: duration desc"`) so the interpretation is transparent and hand-correctable; greys out shot-type/player/score tokens with an inline "not available yet". No LLM, no network call.

60 - **\*\*RallyList/Grid\*\*** ? one card per ``play_segment``: ordinal (by ``start_time``), ``mm:ss`` start, duration (``end?start``), and a shot-count + confidence chip **\*\*only when ``metadata.shot_count`` is present\*\*** (absent on detector paths that don't write it). Checkbox per card; the thumbnail area is a placeholder (no endpoint yet).

61 - **\*\*``useRallySelection()`` hook\*\*** ? owns ``selectedIds:Set<number>``, ``displayOrder``, ``apply(queryResult)``, ``toggle(id)``, and bulk Select-all / Clear / Invert. Unit-tested (matches the exec-plan's "API client + review-state logic" bar).

62 - **\*\*``SelectionFooter`` (sticky)\*\*** ? live "N clips / total M:SS"; a **\*\*``min_shot_count`` warning\*\*** when a selected rally would be dropped at compile (read from ``GET /api/config``); reel-name input; one **\*\*Build\*\*** button.

63 - **\*\*``CompileResult``\*\*** ? ``<video src=GET /api/highlights/...>`` preview of the *compiled reel* + a download link.

64 - **\*\*``TypedApiClient``\*\*** ? ``query``, ``compile``, ``buildDownloadUrl``; the typed seam to the engine.

65

66 **\*\*4.2 Data flow\*\*** ``[verified anchors inline]``

67 Gemini segmenter (process time) writes ``play_segments {id, start/end_time, duration, metadata.shot_count/confidence/detector}`` ``[gemini.py:461-498]``

68 ? SPA ``POST /api/query {video_id}`` ? ``{play_segments:[...]}`` ``[main.py:331-340]`` (**\*\*fix\*\*** the static UI's ``data.rallies`` read ? ``play_segments`` ``[app.js:344]``)

69 ? normalize to ``RallyCard[]`` + one shared ``selectedIds:Set``; ``server/`` ``receiver/`` ``events`` are present in the payload but motion-only fabrications ? **\*\*not surfaced\*\*** ``[motion.py:191-198]``

70 ? all input modes mutate the **\*\*same\*\*** ``selectedIds``: (a) ``parseQuery()`` predicate over duration/ordinal/shot\_count; (b) checkbox toggle + bulk ops; (c) *\*(post-MVP)\** timeline band

71 ? footer recomputes count + ?duration **\*\*client-side\*\***, and warns for any selected rally with ``shot_count < stitching.min_shot_count`` ``[main.py:362-383]``

72 ? **\*\*Build\*\*** ? ``POST /api/compile {video_id, segment_ids: [...selectedIds], output_filename}`` ? stitcher de-dupes via ``_merge_overlapping`` ``[compiler.py:311-317]``, cuts ``time_pairs`` from ``videos.filepath``, re-encodes per clip ? concat ? one MP4 ``[main.py:386-393, compiler.py:303]``

73 ? compile returns a **\*\*server-local filepath, not a URL\*\*** ``[main.py:394]`` ? SPA builds ``GET /api/highlights/{video_id}/{output_filename}`` to preview in ``<video>`` + download ``[main.py:307-328]``.

74

75 **\*\*4.3 Engine API additions (none for MVP; tagged against real gaps)\*\***

76 | Endpoint | Purpose | Status |

77 |---|---|---|

78 | ``POST /api/query`` | List rallies (only path) ? fix SPA to read ``play_segments`` | **\*\*exists\*\*** |

79 | ``POST /api/compile`` | Stitch selected ``segment_ids`` ? MP4 | **\*\*exists\*\*** |

80 | ``GET /api/highlights/{video_id}/{filename}`` | Stream/download compiled reel |

```

**exists** |
81      | `GET /api/config` | Read `stitching.min_shot_count` for the warning | **exists** |
82      | `GET /api/jobs/{job_id}` | Carry `result.video_id` into the picker | **exists** |
83      | `POST /api/compile` ? add `download_url` | Return `/api/highlights/...` so SPA stops
hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **modify**
|
84      | `GET /api/health` | Engine-offline banner ping the SPA is specified to use
(`HOSTING_PLAN.md:42`) | **new** |
85      | `GET /api/videos/{video_id}/rallies` | RESTful/cacheable alias returning the same
`{play_segments}` shape | **new** |
86      | `GET /api/videos` | List a user's videos for a Match picker (only `get_video(id)`
exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
87      | `GET /api/videos/{video_id}/source` (Range) *or* `/clip?start=&end=` | Stream
source/per-span video so a rally previews before selection ? neither exists | **new** |
88      | `GET /api/videos/{video_id}/rallies/{segment_id}/thumbnail` | Per-rally poster still
(no thumbnail column; WASB frames are transient) | **new** |
89      | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?
`candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90
91      **? Dead-ends / do-NOT:** never surface `server`/`receiver`/`events` (motion-only
randoms `[motion.py:191-198]`); never consume shot-type/player/score (no data); never render a
confidence **meter** ? `shot_count_confidence` is a Gemini *string* (often `Unknown`) and
`play_segments` has **no** numeric confidence column.
92
93      ## 5. Risk table
94      | Risk | Severity | Mitigation |
95      |---|---|---|
96      | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the *final* reel;
preview/timeline deferred to M3 behind a new endpoint |
97      | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks
broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]`) |
98      | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-
rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true
scrubber until source streams |
99      | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))`
fallback) `[gemini.py:470-485]` | Med | Label "approx."; confidence shown as the raw string, not
a meter |
100     | No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High
for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate
`GET /api/videos` behind PR-3 scoping |
101     | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for
MVP; `download_url` field (M1) removes the brittleness |
102     | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate
IA: query+grid = one screen; correction = separate ? corpus screen (owner call, $8) |
103
104     ## 6. Honest limits ? what this does NOT do
105     A green SPA build + passing `parseQuery`/selection unit tests prove the **picker logic
only** ? **not** that the engine's rally boundaries are correct (the measured **rally-END**
weakness is a separate quality track, unaffected by this UI). "Conversational" here means a
**deterministic four-dimension parser**, not natural-language understanding. Nothing is
implemented in `web/` yet ? this is `[design]`. The feature curates *what the engine already
found*; it cannot surface a rally the detector missed, nor split/merge a mis-bounded one (that's
the M4 correction loop).
106
107     ## 7. Deliverables & progress tracker ? **source of truth**
108     Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. CI (`npm test`) must stay green; merge to `master`
auto-deploys the site (docs/`web/` changes don't touch the live `site/` Worker).
109
110     | ID | Deliverable | Depends on | Gated? | Status | PR |
111     |---|-----|-----|-----|-----|---|
112     | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? |
[#11](https://github.com/avidullu/khelsutra/pull/11) |

```

```

113 | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to
engine `:8000` | P0 | No | ? | ? |
114 | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) |
P1 | No | ? | ? |
115 | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable
tokens) | P2 | No | ? | ? |
116 | P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ?
| ? |
117 | P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of
the reel | P2 | No | ? | ? |
118 | P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns
`download_url` | ? | No | ? | ? |
119 | P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |
120 | P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen |
P7 | ? owner | ? | ? |
121 | P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8
| ? owner | ? | ? |
122 | P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally
Review | P9 | ? owner | ? | ? |
123
124 **Milestone rollout:** *M0/MVP = P1?P5** (pick?reel on existing API, zero new engine
endpoints) . *M1 = P6?P7** (RESTful + reliability polish) . *M2 = P8** (multi-video home) .
*M3 = P9** (per-rally preview/timeline) . *M4 = P10** (corpus correction loop) . *M5** = swap
`parseQuery()` for localized NL behind the same seam (gated on shot/player data landing).
125
126 ## 8. Open questions ? owner / external
127 **Blocking gates ? ? RESOLVED 2026-06-20 (owner) ? see D8?D11 in §2:**
128 1. ~~Default selection ON vs OFF?~~ ? **ON / curate-by-removal** (D8).
129 2. ~~Is preview-before-select a ship-blocker?~~ ? **No**;; text-metadata MVP, preview is
P9 (D9).
130 3. ~~Job-tethered vs multi-video MVP?~~ ? **Job-tethered** (D10).
131 4. ~~`min_shot_count` override?~~ ? **Warn, don't override** in MVP (D11).
132
133 **Still open (nice-to-knows, non-blocking):**
134 5. May user-facing copy call `parseQuery()` "conversational", given conversational QA is
explicitly deferred (`UI_PROPOSAL.md:66`)? Risk of over-promising NL. *Owner positioning/honesty
call.* (Lean: avoid the word; show the parsed-pill instead.)*
135 6. One screen for query+grid (recommended) vs a separate ? corpus/correction screen for
A4? *Owner positioning call.*
136
137 ## 9. Definition of done
138 - [ ] From a finished job, a user can list rallies, query/select, and download a reel ?
on **existing** endpoints.
139 - [ ] Only honest fields (ordinal / duration / shot_count+confidence-when-present) are
shown; fabricated `server`/`receiver`/`events` are hidden.
140 - [ ] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens,
shows a visible parsed-pill, and is unit-tested.
141 - [ ] Selection hook is unit-tested; the `min_shot_count` floor is surfaced, not hidden.
142 - [ ] `npm run build` (and `npm test`) clean; the static UI's `data.rallies` bug is gone
(or the static UI is superseded by the SPA).
143 - [ ] §7 tracker + this DoD updated; doc archived when P1?P5 land (MVP).
144
145 ## 10. References
146 **Engine code anchors `[verified this session]`:** `backend/main.py:121-123` (config),
`:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile +
`min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135`
(play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475`
(query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485`
(honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198`
(fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe
+ entypoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`
(`default_segmenter: "gemini"`).
147 **Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.

```



```

148
149 ---
150 ### Changelog
151 - 2026-06-20 ? Initial `DRAFT`. Consolidated three candidate designs (Timeline / Grid /
RallyPicker) into the hybrid recommendation via a multi-agent design pass; every engine-fact
anchor re-verified against source before writing.
152 - 2026-06-20 ? Owner **approved** the recommendations; locked D8?D11 (default ON ·
preview not a ship-blocker · job-tethered MVP · warn-don't-override the floor); status `DRAFT` ?
IN PROGRESS`; P0 done.
153

```

## 9. user (2026-06-20T08:59:37.691Z)

```

1      # Portability ? Cloudflare now, always-on VM if needed
2
3      > **Status:** design + proven (2026-06-20). **Decision:** keep the site/form host-
portable so
4      > moving off Cloudflare to an owner-run always-on VM ("if Cloudflare acts out") is an
**adapter
5      > swap, not a rewrite.**
6
7      ## The shape that makes it portable
8      - **Logic is runtime-agnostic.** `src/index.js` uses Web-standard `Request`/`Response`
and pure
9      helpers (`validate`, `toRow`, `isBot`, `handleRequest`) ? no host-specific calls.
10     - **All DB coupling is in adapters** (`src/store.js`): `d1Store` (Cloudflare D1) and
`sqliteStore`
11     (node:sqlite / better-sqlite3). `handleRequest(request, store)` takes the store by
injection.
12     - **One schema, one SQL.** `schema.sql` and the `INSERT` are plain SQLite, so D1 and a
VM's SQLite
13     file are byte-for-byte compatible ? zero dialect change. (Postgres later = minor
placeholder tweaks.)
14
15     ## Two hosts, same code
16     | | Cloudflare (now) | Always-on VM (fallback) |
17     |---|---|---|
18     | Entry | `src/index.js` (Worker `fetch`) | `server.js` (`node:http`) |
19     | Static | `env.ASSETS` | `server.js` serves `public/` |
20     | Storage | `d1Store(env.DB)` ? D1 | `sqliteStore(db)` ? `node:sqlite` file |
21     | Country | `cf-ipcountry` header | optional (GeoIP / null) |
22     | Deploy | git ? Workers Builds | `node server.js` under systemd |
23     | TLS/DNS | Cloudflare | keep Cloudflare in front, **or** Caddy/Let's Encrypt |
24
25     **Enforced, not asserted.** `node --test src/` runs BOTH paths and CI
(`.github/workflows/test.yml`,
26     Node 24) runs them on every PR, so the repo stays compatible with both hosts:
27     - `src/index.test.js` ? the Cloudflare Worker + D1 logic.
28     - `src/server.test.js` ? **boots the real `server.js`** (node:http + node:sqlite) and
round-trips it
29     over HTTP into SQLite (the same proof we run on the droplet, automated).
30
31     For a live check against a real VM: `SSH_TARGET=root@host scripts/droplet-smoke.sh` (scp
+ boot +
32     round-trip + cleanup). Verified live on the owner's droplet (Ubuntu 24.04, isolated
`/srv/khelsutra-site`).
33
34     ## VM deploy runbook ("if Cloudflare acts out")
35     Node 22+ only ? **no `npm install`** (node:sqlite is built in).
36     ```
37     # on the VM:
38     git clone ? && cd khelsutra/site          # or scp the site/ folder
39     PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
40     # put it under systemd (auto-restart) + a reverse proxy (Caddy/nginx) for TLS,
41     # then point DNS at the VM. ~1 hour; logic + schema unchanged.

```

```

42     ``
43
44     ## Honest notes
45     - "Cloudflare acts out" usually means **Workers/D1 pricing or limits, not DNS** ? so the
cheapest
46     move is to keep Cloudflare DNS/CDN in front and shift only the **compute** to the VM
(point a
47     route/origin at the box). Full exit (own DNS + Caddy TLS) is also supported.
48     - You can run **both** (Cloudflare primary + VM warm standby) since the schema is
identical; a
49     periodic `access_requests` export keeps them in sync.
50     - `server.js` is intentionally dependency-free; if you later want Postgres or a
framework (Hono runs
51     on Workers *and* Node), it slots in behind the same `store` + handler.
52

```

## 10. assistant (2026-06-20T08:59:40.847Z)

Now let me explore the directory structures of site/ and web/, and the engine's relevant docs (UI\_PROPOSAL, HOSTING\_PLAN, the storage backend, worker entry point).

## 11. user (2026-06-20T08:59:43.318Z)

```

=== TOP LEVEL ===
total 30
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:19 .
drwxr-xr-x 1 avidu 197609    0 Jun 18 11:14 ..
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:29 .git
-rw-r--r-- 1 avidu 197609  271 Jun 20 13:05 .gitattributes
drwxr-xr-x 1 avidu 197609    0 Jun 20 12:43 .github
-rw-r--r-- 1 avidu 197609   60 Jun 10 23:12 .gitignore
-rw-r--r-- 1 avidu 197609 3467 Jun 20 14:19 NEXT_STEPS.md
-rw-r--r-- 1 avidu 197609  925 Jun 20 08:37 README.md
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:19 docs
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:22 site
drwxr-xr-x 1 avidu 197609    0 Jun 10 23:12 web

```

```

=== site/ ===
site/make_og.py
site/package.json
site/public/capture.html
site/public/index.html
site/public/og.png
site/README.md
site/schema.sql
site/scripts/droplet-smoke.sh
site/scripts/parity-check.sh
site/server.js
site/src/assets.test.js
site/src/deploy-config.test.js
site/src/index.js
site/src/index.test.js
site/src/notify.js
site/src/server.test.js
site/src/store.js
site/wrangler.toml

```

```

=== web/ ===
web/README.md

```

```

=== docs/ ===
PER_RALLY_SELECTION.md
PORTABILITY.md
README.md
REQUEST_ACCESS_FORM.md

```

```
=== .github/workflows ===
parity.yml
test.yml
```

## 12. assistant (2026-06-20T08:59:46.053Z)

Let me read the khelsutra docs/README.md, NEXT\_STEPS.md, REQUEST\_ACCESS\_FORM.md, and the engine's UI/hosting docs plus the storage/worker code anchors.

## 13. user (2026-06-20T08:59:46.545Z)

```
1      # khelsutra docs index
2
3      ## Tracked work (this repo)
4      - **`PER_RALLY_SELECTION.md`** ? interactive pick-rallies ? reel builder (RallyPicker).
The alpha
5      SPA over the engine's rally index; hybrid grid + deterministic query bar on the
**existing**
6      `/api/query` + `/api/compile` contract. `DRAFT` (owner-gated on scope).
7      - **`REQUEST_ACCESS_FORM.md`** ? segmented lead-capture form + Worker/D1 backend
(shipped).
8      - **`PORTABILITY.md`** ? the Node+SQLite VM-host runbook (Cloudflare ? droplet parity).
9
10     ## Engine planning docs
11     Canonical planning docs currently live in the engine repo (`badminton-highlight-
indexer/docs`):
12     `UI_PROPOSAL.md` · `HOSTING_PLAN.md` · `UI_ALPHA_EXECUTION_PLAN.md`. They migrate here
when the
13     control plane lands (per ADR-009 in the engine repo's
`docs/archives/decisions/DECISIONS.md`).
14
```

## 14. user (2026-06-20T08:59:47.117Z)

```
1      # Next steps (live) ? KhelSutra brand / site
2
3      Prioritized "what to do next" for the **`khelsutra` product repo**. Refresh at each
checkpoint.
4
5      **Where we are (2026-06-20):** `khelsutra.guru` is **LIVE and fully shipped** ? landing
page + `/capture`
6      checklist + OG image + a **segmented Request-Access form** (academy/individual + GPU +
storage + free
7      text). On **Cloudflare** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the
form persists to
8      D1 **and emails a summary to the owner on every registration** (Email Routing send
binding ? verified
9      live). **Git auto-deploy is connected** (Workers Builds, root `site/`): merge to
`master` ? deploys;
10     **CI runs `npm test` (20 green) on every PR**. The same form also runs on a dependency-
free **Node +
11     SQLite** host (`site/server.js`), **live + browsable on the D0 droplet ?
http://168.144.126.176:8787**
12     (systemd `khelsutra-site`). PRs #1?#8 merged. Docs: `docs/REQUEST_ACCESS_FORM.md`,
`docs/PORTABILITY.md`.
13
14     ## Next product threads (owner ideas, 2026-06-20)
15     - **Marketing video on the site** ? use Grok/Gemini image-generation to make a short
video (themed for
16     **serious enthusiasts**) walking the whole record ? index ? reel process, and **embed
it** on the
17     landing page. Design-first (storyboard + tone), then produce + embed.
18     - **Interactive per-rally selection** (big product feature) ? the engine detects
rallies; present them
```

```

19      in a UI so the user **picks which rallies go into the highlight** ? dynamic per-video,
**not a fixed
20      threshold/setting**. Pairs with the "ask for a reel" conversational builder already
teased on the
21      site. Lives in the alpha SPA (`web/`) over the rally index the engine produces
(peristed candidate
22      segments). **? Tracked design doc: `docs/PER_RALLY_SELECTION.md` (DRAFT, owner-gated
on scope).**
23      Key finding: the **MVP needs zero new engine endpoints** ? it rides the existing `POST
/api/query`
24      ? select `segment_ids` ? `POST /api/compile` ? `GET /api/highlights`. Owner decisions
pending in
25      the doc's §8 (default selection ON/OFF; is preview-before-select a ship-blocker; job-
tethered vs
26      multi-video MVP; whether the manual selection overrides the silent `min_shot_count`
floor).
27
28      ## Queued (site polish)
29      - **Always-Use-HTTPS + HSTS** (Cloudflare SSL/TLS) ? closes the plain-HTTP flag.
30      - **Cloudflare Web Analytics** ? paste the token into `site/public/index.html` (or
enable auto-setup).
31      - **`/capture` footer link** ? currently `/capture.html` (307?`/capture`); change to
`/capture`.
32      - **Droplet ? production** ? TLS + a subdomain (`vm.khelsutra.guru` via Caddy) if it
should be a real
33      standby; add **VM-path SMTP** so the Node host also emails (currently `noopNotify`).
34      - **DMARC** TXT for deliverability; delete any orphan `khelsutra-site` Worker; clean CF
D1 test rows
35      (`deploy-test@`, `email-test@`). Revoke the droplet SSH key
(`~/ssh/khelsutra_droplet`) when done.
36
37      ## Bigger arcs (design-first)
38      - **Alpha SPA** in `web/` (Vite + React + TS + Tailwind) ? hosts the per-rally selection
UI above.
39      - **BYO-storage + autosync** ? connect Drive/OneDrive/GCS/S3, watch the folder, auto-
prepare reels
40      (engine `PLATFORM_ARCHITECTURE.md §3`). The form's GPU/storage answers are the demand
signal.
41      - **Control plane** (`control-plane/`) at beta.
42
43      ## Convention
44      Substantial work gets a tracked doc in `docs/`; one PR per item off `origin/master`;
merge to `master`
45      auto-deploys (CI must be green).
46

```

## 15. user (2026-06-20T08:59:47.385Z)

```

1      # Request-Access form ? segmented lead capture + setup discovery
2
3      > **Status:** `IN PROGRESS` . **Owner:** Avi . **Created:** 2026-06-20 . **Last
updated:** 2026-06-20
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`
5      > **Relation:** first tracked work item in the `khelsutra` repo; mirrors the engine
repo's
6      > `PROJECT_DOC_TEMPLATE` discipline (status header . decisions-locked . §6 tracker .
DoD).
7      > Feeds the platform direction in the engine repo's `PLATFORM_ARCHITECTURE.md §3`
(StorageBackend /
8      > ComputeBackend) and `VIDEO_LOCALITY_MODEL.md` ? see §7.
9
10     ---
11
12     ## 0. TL;DR
13     Replace the landing page's `mailto:` "Request access" CTA with a real form that (a)

```

```

segments
14    **academies vs individual enthusiasts**, (b) asks the two questions that actually decide
how we'd
15    serve a user ? ***"do you have a GPU machine?"** (compute placement) and ***"where do your
videos
16    live?"** (storage backend) ? and (c) keeps a free-text message. Submissions are stored
in our own
17    **Cloudflare D1** via a small **Worker** (`POST /api/request-access`). The form doubles
as PMF +
18    architecture research.
19
20    ## 1. Problem & goal
21    Today the CTA is `mailto:hello@khelsutra.guru` ? zero structure, nothing collected, no
segmentation.
22    We want **structured, queryable** interest signals from two distinct audiences, captured
the moment
23    someone raises a hand, **without shipping lead data to a third party** (privacy-first
ethos).
24    **Goal:** a clean on-brand form whose responses land in our own Cloudflare account,
queryable later.
25
26    ## 2. Decisions locked
27    - **Backend = Cloudflare Worker + D1** (owner decision, 2026-06-20). Data stays in our
Cloudflare
28    account; queryable; no third party. (Alternatives considered: hosted form
(Tally/Formsprees ?
29    third-party data), Worker?email-only (no DB), structured `mailto` (no collection).
Rejected for
30    the reasons in parens.)
31    - **Two audience segments:** `academy` and `individual` ? the form lightly adapts
(academy also
32    asks club name + #courts).
33    - **The two discovery questions are first-class** (not optional polish): GPU/compute +
storage ?
34    they map 1:1 onto the BYO-compute / BYO-storage architecture forks ($7).
35    - **Free-text message retained** ("Anything else?").
36    - **Privacy:** store only what's submitted + a coarse country (from
`request.cf.country`); **no raw
37    IP**; consent checkbox for "email me about the alpha"; this data is
**operations/contact only**,
38    never a training source (same guardrail as the engine repo).
39    - **Deploy model changes:** the site moves from dashboard direct-upload ? **Wrangler**
(Worker with
40    static assets + D1 binding). Documented in $5; owner runs the `wrangler` deploy (our
account/auth).
41
42    ## 3. Form spec
43    | Field | Type | Notes |
44    |---|---|---|
45    | Segment | radio | `academy` \ | `individual` (required) |
46    | Name | text | required |
47    | Email | email | required (so we can reply) |
48    | City | text | optional |
49    | Club / academy name | text | shown for `academy` |
50    | Courts recorded | select | `1` \ | `2-4` \ | `5+` (academy; optional) |
51    | **GPU machine for local processing?** | radio | `yes` \ | `unsure` \ | `no` \ | `prefer-
cloud` |
52    | **Where do your recordings live / how would you share them?** | checkboxes (multi) |
`gdrive` \ | `onedrive` \ | `dropbox` \ | `bucket` (S3/GCS) \ | `local` (SD/drive) \ | `unsure` |
53    | Message | textarea | free text ("Anything else?") |
54    | Consent | checkbox | "Okay to email me about the KhelSutra alpha" |
55    | `hp` | hidden honeypot | anti-spam; if filled ? silently drop |
56
57    ## 4. Design / architecture
58    - **Frontend** (`site/index.html`): the header + hero "Request access" buttons scroll to

```

```

the
59     `#join` form; `#join` becomes the form. On submit, JS `POST`s JSON to `/api/request-
access`.
60     **Progressive enhancement:** if the endpoint isn't live yet (pre-Worker), it falls
back to a
61     prefilled `mailto:hello@` so the form is useful from the first re-upload.
62     - **Worker** (`/api/request-access`): validates (email present, honeypot empty), inserts
into D1,
63     returns `200 {ok:true}`; serves the static assets for all other routes.
64     - **D1 schema** (`access_requests`): `id, created_at, segment, name, email, city,
org_name, courts,
65     has_gpu, storage (csv), message, consent, country, user_agent`.
66     - **Anti-spam:** honeypot first; add **Cloudflare Turnstile** in P2 if spam appears.
67     - **Notify (P2):** optional email ping to `hello@` per submission (MailChannels /
Cloudflare Email).
68
69     ## 5. Deploy (owner-run; our Cloudflare account)
70     ```
71     wrangler d1 create khelsutra-access          # once; copy the database_id into
wrangler.toml
72     wrangler d1 execute khelsutra-access --file=site/schema.sql # create the table
73     wrangler deploy                               # ships the Worker + static assets + D1
binding
74     ```
75     > Until the Worker ships, the form works via the `mailto` fallback (re-upload `site/` as
today).
76     > Git?Cloudflare auto-deploy still intentionally not connected.
77
78     ## 6. Deliverables & progress tracker ? source of truth
79     Legend: ? Todo · ? In progress · ? Done. One PR per row off `origin/master`.
80     | ID | Deliverable | Gated? | Status | PR |
81     |----|-----|-----|-----|----|
82     | P0 | This design doc + `NEXT_STEPS.md` | No | ? | #2 |
83     | P1 | Form UI (segments + GPU + storage + free text + consent) w/ `mailto` fallback |
No | ? | #2 |
84     | P2 | Worker + D1 + `wrangler.toml` + `schema.sql` + **unit tests (10 green)** | owner
deploy | ? | this PR (built+tested; owner runs `wrangler d1 create` + `deploy`) |
85     | P3 | Anti-spam (Turnstile) + per-submission email notify | No | ? | ? |
86     | P4 | Tiny admin/export view (auth'd) to read responses | owner | ? | ? |
87
88     ## 7. Why this matters beyond lead capture
89     The two questions are **the architecture forks** from the engine repo's BYO analysis:
90     - **GPU?** ? compute placement: desktop/local (their GPU) vs hosted vs zero-infra-BYO
91       (`PLATFORM_ARCHITECTURE.md §3.2`, `DESKTOP_APP_PLAN.md` P0 spike).
92     - **Storage?** ? which `StorageBackend` + whether autosync is worth building
93       (`PLATFORM_ARCHITECTURE.md §3.1`, the cloud-storage/autosync thread).
94     So early responses directly inform what to build next ? not just who to email.
95
96     ## 8. Open questions
97     1. Turnstile now or wait for spam? (Default: honeypot now, Turnstile if needed.)
98     2. Email-notify per submission, or just query D1 weekly? (Default: notify in P2.)
99     3. Do we localize the form (Kannada/Hindi) for academies now or at beta? (Default:
English now.)
100
101     ## 9. Definition of done
102     - [ ] Form replaces the `mailto` CTA, segments academy/individual, asks GPU + storage +
free text.
103     - [ ] Submissions persist to Cloudflare D1 via the Worker; honeypot drops bots.
104     - [ ] Privacy posture honored (no raw IP; consent; contact-only).
105     - [ ] Deployed live on khelsutra.guru; a test submission appears in D1.
106     - [ ] Doc tracker + DoD updated; archived when P1?P3 land.
107

```

Now let me look at the engine-side docs referenced (UI\_PROPOSAL, HOSTING\_PLAN) and the storage/worker code anchors to ground the appliance/operator console relation.

### 17. user (2026-06-20T08:59:56.556Z)

Already up to date.

```
=== docs UI/HOSTING ===
```

```
DECOUPLED_COMPUTE
```

```
DESKTOP_APP_PLAN.md
```

```
HOSTING_PLAN.md
```

```
NEXT_STEPS.md
```

```
PLATFORM_ARCHITECTURE.md
```

```
UI_ALPHA_EXECUTION_PLAN.md
```

```
UI_PROPOSAL.md
```

```
VIDEO_LOCALITY_MODEL.md
```

```
=== storage/ ===
```

```
backend/storage/base.py
```

```
backend/storage/gcs.py
```

```
backend/storage/gdrive.py
```

```
backend/storage/local.py
```

```
backend/storage/ref.py
```

```
backend/storage/s3.py
```

```
backend/storage/__init__.py
```

```
backend/storage/__pycache__/base.cpython-313.pyc
```

```
backend/storage/__pycache__/gcs.cpython-313.pyc
```

```
backend/storage/__pycache__/gdrive.cpython-313.pyc
```

```
backend/storage/__pycache__/local.cpython-313.pyc
```

```
backend/storage/__pycache__/ref.cpython-313.pyc
```

```
backend/storage/__pycache__/s3.cpython-313.pyc
```

```
backend/storage/__pycache__/__init__.cpython-313.pyc
```

```
=== worker/ ===
```

```
backend/worker/__init__.py
```

```
backend/worker/__main__.py
```

```
backend/worker/__pycache__/__init__.cpython-313.pyc
```

```
backend/worker/__pycache__/__main__.cpython-313.pyc
```

### 18. assistant (2026-06-20T09:00:00.362Z)

The engine repo cwd is the badminton-highlight-indexer (the Bash tool resets cwd). Let me read the key engine docs and code for the operator-console grounding: UI\_PROPOSAL, HOSTING\_PLAN, the worker entry, storage base, and the DECOUPLED docs dir.

### 19. user (2026-06-20T09:00:01.313Z)

```
1      # UI Proposal ? khelsutra.guru web app (alpha ? beta)
2
3      **Status:** PROPOSAL ? awaiting owner ratification (no implementation starts without
explicit approval, per CLAUDE.md).
4      **Date:** 2026-06-10 . **Inputs:** PMF_MARKET_RESEARCH, COMMERCIALIZATION,
PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES,
reporting/spec.yaml, current API + static UI.
5      **Companion:** `docs/HOSTING_PLAN.md` (domain, topology, costs).
6
7      ## 1. Strategic frame ? why a UI now, and what it secretly is
8
9      The UI is not just product surface; it is the corpus-growth engine. The single
highest-leverage
10     item in NEXT_STEPS is growing the golden corpus, and the v1 scope already commits to a
11     human correction loop. Every tester who accepts/rejects/nudges a rally boundary in
the review
12     screen produces exactly the labels the Gen-0/M0.x track is starved for (subject to the
per-user
13     training opt-in guardrail). 10?25 alpha testers ? a steady stream of labeled, real-
world,
```

```

14 multi-court-biased footage ? the distribution where the owned model's 0.66 ship target
lives.
15
16 Serving economics per the 2026-06-10 quality table: alpha serves the **Gemini path**
17 (0.93 on clean footage, ~$0.05/run) ? no GPU required in the request path. The
WASB/owned-model
18 path stays on the owner's box and is *not* exposed in the alpha UI (segmenter choice
hidden).
19
20 **The CLI is not replaced.** The web app talks to the same FastAPI surface the CLI
drives; the CLI
21 remains the power-user/ops tool.
22
23 ## 2. Personas (from PMF research)
24
25 - **Coach / academy (primary, B2B, India):** batches of match recordings, wants rally
review,
26 per-player reels for students/parents, per-rally stats. Already records on static
cameras.
27 - **Enthusiast (B2C funnel):** records own matches, wants a highlight reel + basic
stats, shares
28 with friends. Tripod discipline is the known adoption risk ? the UI must teach
recording
29 guidelines *before* upload, not fail videos after.
30
31 ## 3. ALPHA ? "quick & dirty" (goal: 10?25 invited testers, end-to-end, zero hand-
holding)
32
33 **Stack:** React + Vite + Tailwind SPA, deployed per HOSTING_PLAN (static hosting, $0).
34 API = existing FastAPI on the GPU box via tunnel at `api.khelsutra.guru`.
35 **Auth:** Cloudflare Access email-allowlist (OTP), free ?50 users, **zero auth code** ?
backend
36 reads the authenticated-email header for per-user scoping (alpha-grade tenancy: header
is
37 trustworthy because the tunnel only accepts Access-authenticated traffic).
38
39 ### Alpha screens (6)
40
41 | # | Screen | What it does | Backed by |
42 |---|-----|-----|-----|
43 | A1 | **Matches** (home) | User's videos, status chip per video (verdict ?/??/?),
upload CTA | `videos` + owner scoping (B4) |
44 | A2 | **Upload & Process** | Drag-drop; inline recording-guidelines checklist (pre-
upload education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing |
B2 upload endpoint ? B1 job |
45 | A3 | **Job progress** | Stage list (validation ? segmentation ? AI confirm ? report)
with live status; verdict banner on completion | B1 `GET /api/jobs/{id}` poll |
46 | A4 | **Rally review** ? | Video player + rally list; click rally ? seek & play that
span; per-rally **Accept / Reject / nudge start?end ±0.5 s**; bulk "all good"; chips: duration,
shot count (+confidence) | B6 correction endpoints ? `candidate_segments.human_label/notes` ?
**the corpus engine** |
47 | A5 | **Highlights** | Select accepted rallies ? name reel ? compile ? download; list
of past reels | `/api/compile` + B3 download endpoint |
48 | A6 | **Report** | Renders the customer-audience report (verdict, notes, FAQ refs);
"nerd view" toggle shows technical.md | existing reporting artifacts |
49
50 ### Backend work the alpha actually requires (honest dependency list)
51
52 | ID | Item | Notes |
53 |----|-----|-----|
54 | B1 | **Async job model ? DEFERRED_HARDENING_PLAN #16** | The committed next PLATFORM
slice; `POST /api/process` ? 202 + `job_id`, `jobs` table (user_version 1?2),
ThreadPoolExecutor, `GET /api/jobs/{id}`. **Approving this proposal = greenlighting #16.** A
5?60 min blocking HTTP call cannot sit behind a tunnel UI. |
55 | B2 | **Chunked upload endpoint** | `POST /api/upload` accepting ?90 MB parts (free-

```



tier proxy body limit is 100 MB), assemble ? MD5 ? register under `allowed\_video\_root/<user>/`. Size cap (e.g. 4 GB) + mp4/mov allowlist. |

56 | B3 | **\*\*Download endpoint\*\*** | `GET /api/highlights/{video\_id}/{name}` streaming the compiled reel (today `/api/compile` returns a server-local path the browser can't reach). |

57 | B4 | **\*\*Per-user scoping\*\*** | `owner\_email` column on `videos` (migration), read from the Access header, filter every query. Alpha-grade, single DB. |

58 | B5 | **\*\*Retire/replace static UI\*\*** | The SPA supersedes `backend/api/static` (which has a known `rallies` vs `play\_segments` field bug ? don't fix, replace). |

59 | B6 | **\*\*Correction-loop endpoints\*\*** | `PATCH /api/segments/{id}` ? accept/reject/boundary-nudge persisted to `candidate\_segments.human\_label`/`notes` (+ provenance: who, when). This is v1-scope "correction loop" made real. |

60 | B7 | **\*\*Exposure hardening\*\*** | CORS narrowed to the app origin; basic rate limit; upload quota per user; `allowed\_video\_root` set; Gemini key rotated (already in NEXT\_STEPS housekeeping). |

61

62 ### Alpha non-goals (explicit)

63

64 No payments, no orgs/teams, no athlete profiles, no share links, no owned-model serving, no

65 multi-sport selector (badminton only), no notifications, no mobile-perfect layouts (usable on

66 phone, optimized for laptop), no conversational QA (deferred in roadmap).

67

68 ### Alpha effort estimate

69

70 B1 ? 2?3 sessions · B2?B7 ? 2?3 sessions · SPA (6 screens) ? 3?4 sessions ? **\*\*~8?10**

working

71 sessions\*\* to first tester invite. Sequencing: B1 ? B2/B3/B4 ? SPA skeleton + A2/A3 ? A4 ? A5/A6 ? B7 ? invite.

72

73 ## 4. BETA ? polished (goal: paying pilot academies)

74

75 Maps deliberately onto PLATFORM\_ARCHITECTURE P1?P3 ? the UI roadmap and the platform roadmap are

76 the same roadmap.

77

78 1. **\*\*Drop 1 ? Accounts & academies (P1):\*\*** real auth (Firebase Auth ? GCP-native, free tier ? or

79 Clerk), academy workspace, coach invites athletes (**\*\*18+ only\*\***, per guardrail), roles

80 (owner/coach/athlete). Coach?athlete stays an **\*\*optional overlay\*\*** on the generic schema.

81 2. **\*\*Drop 2 ? Storage & worker (P2/P3):\*\*** direct-to-cloud resumable uploads (signed URLs ? bypasses

82 proxy body limits entirely), reels served from object storage; control plane moves to Cloud Run;

83 **\*\*GPU box becomes the first pull-based `byo\_worker`\*\*** (outbound-only, exactly P3's security

84 model). Email notification when a reel is ready.

85 3. **\*\*Drop 3 ? Review v2:\*\*** keyboard shortcuts (J/K/L, [/] nudge), frame-step, timeline minimap with

86 rally bars, batch accept, rally tags, athlete-facing **\*\*expiring share links\*\***, per-player

87 aggregates and match-over-match trends (v1 stats only ? **\*\*no gait/biometrics\*\***, GDPR Art. 9).

88 4. **\*\*Drop 4 ? Monetization & consent:\*\*** Razorpay per-match credits (INR; PMF: B2C \$60?180/yr,

89 academy \$300?1,500/yr equivalents, tuned per-match for India), free-tier match allowance;

90 **\*\*training-data opt-in\*\***: per-user toggle default OFF + per-match consent record + adult-only

91 attestation at upload (guardrails: minors excluded from training pool; separate explicit opt-in).

92

93 Beta effort: ~4?6 weeks part-time across the four drops; each drop independently

```

shippable.
94
95     ## 5. Decisions needed from owner
96
97     1. **Greenlight** alpha scope above (= greenlights DEFERRED_HARDENING_PLAN #16)?
98     2. **RESOLVED 2026-06-10:** product name = **KhelSutra Guru**; domain =
**khelsutra.guru** (GoDaddy receipt, 3-yr registration + domain protection, renews 2029-06).
99     3. Per-user **training opt-in default OFF** ? confirm (recommended; conservative per
guardrails).
100    4. Alpha invite list: the C13 interview academies + camera-pilot contacts?
101    5. SPA stack confirm: React + Vite + Tailwind (proposed) vs anything you prefer.
102    6. **RESOLVED 2026-06-10:** separate **private repo `khelsutra`** (owner-ratified ? the
product's
103        "heart and brain"): SPA in `web/`, control plane joins at beta; backend PRs B1?B7
stay in the
104        indexer repo. Keeps the indexer's licensing/test discipline untouched.
105
106    *Once ratified, decisions 1?6 get recorded in `docs/archives/decisions/DECISIONS.md` per
convention.*
107

```

## 20. user (2026-06-20T09:00:01.608Z)

```

1      # Hosting Plan ? khelsutra.guru
2
3      **Status:** PROPOSAL ? awaiting owner ratification.
4      **Date:** 2026-06-10 . **Constraints:** budget $10?25/mo . GCP credits available .
DigitalOcean
5      acceptable . alpha backend = owner's GPU box behind a tunnel . 10?25 alpha testers .
registrar = GoDaddy.
6      **Companion:** `docs/UI_PROPOSAL.md`.
7
8      ## 1. Recommendation in one line
9
10     **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
11     **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**
(Cloud Run +
12     GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
13
14     Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
15     Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
16     exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a
home box with
17     a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
18     Credits are better spent at beta on compute/storage, which is where GCP is excellent.
19
20     ## 2. Alpha topology ($0/mo)
21
22     ...
23     tester browser
24     ? https
25     ?
26     Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
27     ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
28     ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on
the box)
29
30     ?
31     FastAPI :8000 on the GPU box (Windows)
32     ? Gemini segmenter path (no GPU in request path)
33     ? async jobs (#16), SQLite WAL, output/ on disk

```

```

34     ``
35
36     - **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP
never exposed.
37     - **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
38     `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-
user scoping.
39     - **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in
?90 MB
40     chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
41     - **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly
"processing engine
42     offline" banner (a `/api/health` ping). Beta removes this with the queue.
43     - **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost
(~$0.05/match) and
44     domain renewal.
45
46     ## 3. Beta topology (GCP credits; ~$15?20/mo post-credits)
47
48     ``
49     browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
50                                     ??? api.khelsutra.guru ??? Cloud Run
(FastAPI control plane, scale-to-zero)
51                                     ? jobs
table (Cloud SQL Postgres)
52                                     ? signed
resumable URLs
53                                     ?
54                                     GCS buckets
(uploads / reels, lifecycle ? cold)
55                                     ? pull
jobs, outbound-only
56                                     GPU box = first
`byo_worker` (PLATFORM P3)
57     ``
58
59     - Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit,
no home
60     bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs),
Cloud Run
61     handles Gemini-path jobs entirely in-cloud.
62     - Auth graduates to **Firebase Auth** (free; Google + email sign-in; India-friendly
phone OTP
63     later); Access allowlist retires.
64     - Post-credits estimate: Cloud Run ~$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-
to-zero) .
65     GCS 100 GB ? $2.60 + egress $0.12/GB (reel downloads ? watch this line) . Cloud SQL
smallest
66     ? $10 ? **~$15?20/mo**, inside budget. Egress lever if it grows: serve reels from
Cloudflare R2
67     (zero egress) while keeping everything else GCP.
68
69     ## 4. Alternatives considered
70
71     | Option | Monthly (alpha ? beta) | Pros | Cons |
72     |---|---|---|---|
73     | **Recommended hybrid** (CF edge + box ? +GCP) | **$0 ? ~$0 w/ credits** (~$15?20
after) | Cheapest at every phase; credits used where they matter; maps to PLATFORM P2/P3 | Two
vendors (CF + GCP) |
74     | **All-DigitalOcean** | ~$11 ? ~$17 ($6 droplet + Spaces $5: 250 GiB + **1 TiB egress
included**) | One simple vendor; Spaces' included egress is the best video-download deal; per-
second billing | No credits; no scale-to-zero; still needs a tunnel-equivalent to reach the box
(run `frp`/WireGuard on the droplet) |
75     | **All-GCP from day 1** | ~$18+ ? credits | Single vendor, credits immediately | LB+IAP
standing cost just to gate an alpha; no free tunnel to the home box; credits burn on plumbing |

```

```

76      | **All-Cloudflare** (Pages+Workers+R2) | ~$0 ? ~$5 | Zero egress on R2; cheapest
forever | FastAPI doesn't run on Workers (API stays on box/Cloud Run anyway); ignores GCP
credits |
77
78      ## 5. Domain setup ? GoDaddy ? Cloudflare DNS (one-time, ~15 min, you)
79
80      1. Create a free Cloudflare account ? **Add a domain** ? `khelsutra.guru` ? Free plan.
Cloudflare
81          imports existing records and shows **two nameservers** (e.g.
`xxx.ns.cloudflare.com`).
82      2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ? Change ?
83          "I'll use my own nameservers"** ? paste the two Cloudflare nameservers ? Save.
Propagation:
84          minutes to 24 h. (Registration/renewal stays at GoDaddy; only DNS moves. Optional
later:
85          transfer the registration to Cloudflare Registrar for at-cost .guru renewals.)
86      3. Records (I manage after takeover): `app` ? Pages project; `api` ? `**Cloudflare API token** (you create: dash ? My Profile ? API Tokens) | Zone:DNS>Edit,
Account:Pages>Edit, Account:Access:Apps>Edit, Account:Tunnel>Edit ? scoped to this zone | DNS
records, Pages deploys (wrangler), Access policies, tunnel config |
94      | **GitHub repo** `khelsutra` (private) | Pages connects once (dashboard, you; build
root = `web/`) ? every push auto-deploys | I write the SPA in your Projects folder; commits
pushed = deployed |
95      | **On the box** (one-time, I hand you exact commands) | `cloudflared` installed as a
Windows service + `config.yml` | The tunnel itself |
96      | **GCP (beta only)** | A `khelsutra` project + service account (Cloud Run admin,
Storage admin, Cloud SQL admin) | Control-plane deploys |
97
98      Day-to-day flow after setup: I build ? commit ? push ? Pages deploys the SPA
automatically;
99      API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never
live in the
100     repo; `.env` on the box + Cloudflare/GCP dashboards.
101
102     ## 7. Go-live security checklist (alpha)
103
104     Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?
rate
105     limit + per-user upload quota ? `security.allowed_video_root` set ? upload size cap +
extension
106     allowlist ? **rotate the Gemini API key** (NEXT_STEPS housekeeping ? it was shared in
chat) ?
107     confirm no router port-forwards ? weekly backup: SQLite DB + `output/` ? GCS bucket
(rclone
108     scheduled task; credits).
109
110     ## 8. Decisions needed from owner
111
112     1. Ratify the **hybrid recommendation** (CF free edge now, GCP credits at beta)? Or
force
113         single-vendor (DO / GCP) despite the cost notes above?
114     2. OK to move **DNS to Cloudflare** (registration stays at GoDaddy)?
115     3. Beta reel storage: **GCS (credits)** vs **R2 (zero egress)** vs **DO Spaces (1 TiB
included)** ?
116         can defer until beta Drop 2.
117     4. Approve the access grants in §6 so I can run the setup end-to-end.
118
119     *Ratified choices get recorded in `docs/archives/decisions/DECISIONS.md`.*

```

**21. assistant (2026-06-20T09:00:05.437Z)**

Now let me look at the DECOUPLED\_COMPUTE docs dir and the worker entry / storage base to ground the appliance + operator-console concept against what's actually shipped.

**22. user (2026-06-20T09:00:07.933Z)**

```
=== DECOUPLED_COMPUTE dir ===
docs/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md
docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md
docs/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md
docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md
docs/DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md
docs/DECOUPLED_COMPUTE/06-RISKS-AND-OPEN-QUESTIONS.md
docs/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md
docs/DECOUPLED_COMPUTE/HANDOFF.md
docs/DECOUPLED_COMPUTE/README.md
=== HANDOFF? ===
DECOUPLED_COMPUTE
HARDENING_LOOP_HANDOFF.md
=== worker __main__ head ===
"""Worker dispatcher ? pull from storage ? run the pipeline ? push outputs back.
```

```
python -m backend.worker infer --video-uri <uri> --out-uri <prefix> \
    [--segmenter wasb_hybrid] [--config config.json] [--set k=v ...] [--scratch DIR]
python -m backend.worker train ...          # Gen-0 train-from-bucket (follow-up)
```

The URI scheme selects the storage backend via ``backend.storage`` (a bare path / ``local://`` is an identity passthrough; ``s3://`` / ``gcs://`` is an object store), so existing local runs are byte-for-byte unchanged. Combine with ``RALLY\_DETECTOR\_IMPL=stub`` and ``--set indexing.skip\_ai\_handoff=true`` for a \*\*secret-free, GPU-free\*\* run (no Gemini key) ? the path validated end-to-end on DigitalOcean (``docs/SETUP\_NEW\_MACHINE.md``).

This is a thin orchestrator: it composes the SAME building blocks the ``backend.main`` CLI uses (validator registry ? segmenter ? report), never a forked pipeline.

```
"""
from __future__ import annotations

import argparse
import csv
import json
import os
import sys
import tempfile
from typing import Any, List

from backend import storage
from backend.config import load_config
from backend.infrastructure.database import Database
from backend.pipeline.segmenters.base import segmenter_registry
from backend.pipeline.validators.base import registry as validator_registry
from backend.reporting import emit_indexer_report
from backend.results import Failure
from backend.utils.hashing import compute_video_id

def _coerce(v: str) -> Any:
    """Coerce a ``--set`` string value to bool/int/float where it round-trips, else str."""
    low = v.lower()
    if low in ("true", "false"):
        return low == "true"
    for cast in (int, float):
        try:
```

```

        return cast(v)
    except ValueError:
        pass
    return v

```

```

def _apply_overrides(config: Any, sets: List[str]) -> None:
    """Apply ``a.b.c=value`` overrides onto the typed config (e.g.
    indexing.skip_ai_handoff=true)."""
    for kv in sets or []:
        key, sep, val = kv.partition("=")
        if not sep:
            raise ValueError(f"--set expects k=v, got {kv!r}")
        parts = key.split(".")
        obj = config
        for p in parts[:-1]:
            obj = getattr(obj, p)
        setattr(obj, parts[-1], _coerce(val))

```

### 23. assistant (2026-06-20T09:00:10.431Z)

Let me read the DECOUPLED\_COMPUTE README and HANDOFF for the appliance/operator-console framing, plus the storage base for the reuse seam.

### 24. user (2026-06-20T09:00:11.930Z)

```

1      # Decoupled Compute ? Cloud-Native GPU for Train + Inference
2
3      **Status:** ? PROPOSAL / research ? for owner review. **No code yet.**
4      **Date:** 2026-06-20 . **Owner:** avidullu
5      **Lifecycle:** *Critical active project.* When delivered, **move this whole directory to
`docs/archives/past_projects/`** (keeps the strategy tree clean).
6
7      > **What this directory is.** A single, unified place that answers one question: *can we
lift the
8      > rally-segmentation pipeline off the owner's local Windows/WSL box and run it on a
**rented,
9      > credit-budgeted, cloud-native GPU** ? driven by an API key ? that does **both** (a)
**retrain** the
10     > rally model from a **bucket of videos** and (b) run **inference on new uploads** ? and
**bill each
11     > video / each training run exactly**, including for a non-owner user?* Short answer:
the *design*
12     > already exists across our strategy docs; the *build* (a container/VM image, a real
bucket, a chosen
13     > provider, a cost ledger) does not. This directory inventories what's planned, maps the
real compute,
14     > compares platforms, and lays out a cheapest-first build & test plan that fits the
**$200** budget.
15
16     ---
17
18     ## North Star
19
20     A **decoupled** pipeline where the **control plane** (a thin API that accepts jobs and
reports status)
21     is separated from the **data/compute plane** (the heavy bytes + GPU work), which runs in
a **rented
22     cloud GPU** reading from and writing to an **object-store bucket**. Two verbs:
23
24     - **`train`** ? consume a bucket of videos + labels ? produce versioned model weights
back to the bucket.
25     - **`infer`** ? a new uploaded video ? rally segments + highlight reel + an eval report
back to the bucket.
26

```

```

27     ?with **exact per-video (inference) and per-run (training) cost attribution** so the
cost can be
28     charged to whoever ran the job.
29
30     ---
31
32     ## TL;DR (read this, then dive into the numbered docs)
33
34     1. **Yes ? decoupling is already designed, in depth.**
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md)
35     is a control-plane/data-plane split with a `ComputeBackend` registry (`local | hosted
| byo_worker`)
36     and a `StorageBackend` registry (S3/GCS/Azure via `fsspec`);
[HOSTING_PLAN.md](../HOSTING_PLAN.md)
37     already names **GPU box = first `byo_worker`** with GCS buckets + signed-URL uploads;
$5A specifies
38     **training on a rented GPU** via `ms-swift`. ? **[01](01-EXISTING-PLANS-AND-
GAP.md)**.
39     2. **What's missing is everything that turns those seams into the North Star:** a
container/VM image,
40     a **chosen** provider + an **API-key/provisioning** flow, a **real bucket** (today
it's OneDrive
41     auto-sync ? a proto-bucket), a **download-to-local** shim, path de-hardcoding, a
**bucket-triggered
42     retrain**, a **remote inference endpoint**, and **cost metering**. ?
**[01](01-EXISTING-PLANS-AND-GAP.md)**, **[02](02-COMPUTE-MAP-AND-CONTRACT.md)**.
43     3. **$200 is comfortable ? for the *right* scope.** It easily covers the **inference
pipeline + Gen-0
44     head training** (numpy-class, trains in minutes on a small GPU) decoupled onto one
cheap GPU. It does
45     **NOT** cover the **Gen-2 VLM fine-tune** (ms-swift/QLoRA, multi-GPU, **the largest
open risk,
46     currently unquantified") ? that stays explicitly **out of scope** for this budget. ?
**[03](03-PLATFORM-OPTIONS.md)**, **[06](06-RISKS-AND-OPEN-QUESTIONS.md)**.
47     4. **Docker is optional.** Serverless/Cloud Run paths need an image; a **DO Droplet** or
**SkyPilot**
48     path can run the repo in a plain `venv`/`conda` with no container. We recommend a
container **where you
49     go serverless* and a VM/SkyPilot path otherwise. ? **[03](03-PLATFORM-OPTIONS.md)**.
50     5. **Per-video billing falls out of a serverless per-invocation metering model + a cost
ledger on the
51     already-shipped `jobs` table.** ? **[05](05-COST-ATTRIBUTION.md)**.
52     6. **Cheapest-first: M0 costs $0** ? a CPU container/venv runs the **committed
synthetic, video-free
53     regression fixtures** ([`golden_fixtures.py`](../backend/eval/golden_fixtures.py))
in CI, no GPU,
54     no WSL. Only spend a dollar at M1. ? **[04](04-BUILD-AND-TEST-PLAN.md)**.
55     7. **The WSL detectors run seamlessly off the Windows box ? and we prove it on
DigitalOcean.** The
56     `DetectorRunner` ABC is already built for the swap (WSL plumbing is an isolated
mixin; the `Frame,
57     Visibility,X,Y` CSV is the parity contract); a DO GPU is Ubuntu+CUDA = your WSL conda
env, **minus
58     WSL**. A dedicated **M3.5 parity test on DO** is the guarantee, and it doubles as the
59     `DESKTOP_APP_PLAN` P0 spike. ? **[02](02-COMPUTE-MAP-AND-CONTRACT.md)**,
**[04](04-BUILD-AND-TEST-PLAN.md)**, **[06](06-RISKS-AND-OPEN-QUESTIONS.md)**.
60
61     ---
62
63     ## The budget tension you must decide (see [03](03-PLATFORM-OPTIONS.md) + [06](06-RISKS-
AND-OPEN-QUESTIONS.md))
64
65     You have **$200 in DigitalOcean**, and you're open to **GCS + a Cloud-Run-like**
architecture. These pull
66     in different directions for the **side goal** (exact per-video billing):

```

```

67
68     - **DigitalOcean** has the credits + **Spaces** (S3-compatible, **1 TiB egress
included** ? the best
69     video-download deal), but **no true serverless GPU**, so per-video metering is coarser
(you bill
70     wall-clock on a started/stopped GPU Droplet).
71     - **GCP Cloud Run (GPU) / RunPod Serverless** give **scale-to-zero + exact per-
invocation GPU-seconds**
72     ? which *is* your per-video bill ? but the $200 isn't DO credit there (GCP's
$300/90-day trial is the
73     clean free tier; RunPod's "free credit" is small).
74
75     **Recommended split:** prototype M0?M2 on **DO (use the credits) + DO Spaces**; stand up
the M3 inference
76     *endpoint* where per-invocation metering is native (Cloud Run GPU / RunPod Serverless).
Final call is
77     owner's ? tracked in [06](06-RISKS-AND-OPEN-QUESTIONS.md).
78
79     ---
80
81     ## Contents
82
83     | # | Doc | What it answers |
84     |---|---|---|
85     | **00** | [**HANDOFF ? resume here**](HANDOFF.md) | **? Current status (CPU e2e
working) + the next build (M3.5 native WASB). Read first.** |
86     | 01 | [Existing plans & the gap](01-EXISTING-PLANS-AND-GAP.md) | What we've already
planned for decoupling, and precisely what's unbuilt. |
87     | 02 | [Compute map & data contract](02-COMPUTE-MAP-AND-CONTRACT.md) | Where
GPU/CPU/WSL/API actually live in the 3 repos; what must move to a bucket. |
88     | 03 | [Platform options](03-PLATFORM-OPTIONS.md) | DO vs GCP vs serverless GPU; Docker-
optional alternatives; provider matrix. |
89     | 04 | [Build & test plan](04-BUILD-AND-TEST-PLAN.md) | The two entrypoints, bucket
layout, the M0?M4 milestone ladder, first PR. |
90     | 05 | [Cost attribution](05-COST-ATTRIBUTION.md) | Per-video inference billing + per-
run training cost segmentation for any user. |
91     | 06 | [Risks & open questions](06-RISKS-AND-OPEN-QUESTIONS.md) | The legal/scope/budget
gates and the decisions only the owner can make. |
92     | 07 | [Compute abstraction](07-COMPUTE-ABSTRACTION.md) | How we segment GPU vs CPU work
+ a spec-aware `DeviceContext`/`DeviceProfile` layer (the "GPU Runner"). |
93
94     ---
95
96     ## Verification status (checked against code 2026-06-20)
97
98     These load-bearing facts were confirmed against the current tree (not just the older
strategy docs):
99
100    - ? **Async-job control plane shipped** ? `jobs` table at
[database.py:188](../backend/infrastructure/database.py), worker at
[api/job_worker.py](../backend/api/job_worker.py), `/api/process` ? 202 + `job_id`.
101    - ? **`jobs` table has no cost/tenant columns yet** (cols: id, video_path, segmenter,
player_pool, max_frames, status, policy, result, progress, timings) ? so the cost-ledger delta
in [05](05-COST-ATTRIBUTION.md) is net-new.
102    - ? **Gen-0 harness exists** ?
[training/gen0/harness.py](../training/gen0/harness.py) (one-command train?eval?gate).
103    - ? **Committed synthetic, video-free regression fixtures exist** ?
[backend/eval/golden_fixtures.py](../backend/eval/golden_fixtures.py) ? M0 is a genuine $0
CI gate.
104    - ? **`"output"` path still hardcoded** ?
[trajectory_hybrid.py:196,272](../backend/pipeline/segmenters/trajectory_hybrid.py) (a real
bucket-relocation blocker).
105    - ? **No Dockerfile anywhere** in any of the three repos ? the container is greenfield.
106    - ? **Collector keys by human `video_id`, not MD5** ? `curate.py`; the export manifest
omits an `md5` field, which the MD5-keyed consumer needs (see [02](02-COMPUTE-MAP-AND-

```



```

CONTRACT.md)).
107
108     ## Provenance
109
110     Built from a 13-agent research workflow (?970k tokens, 5 plan-harvest + 3 compute-map +
111     2 cloud-provider
112     agents, then synthesis + an adversarial completeness critic), cross-checked against the
113     owner's strategy
114     docs and the live code. Cost/price figures are **approximate ? verify current provider
115     terms before
116     committing budget.**
117
118

```

## 25. user (2026-06-20T09:00:15.562Z)

```

1     # DECOUPLED_COMPUTE ? Session Handoff / Resume-Here
2
3     **Updated:** 2026-06-20 . **Read this first to resume.** Companion: this whole directory
4     + `docs/SETUP_NEW_MACHINE.md`.
5
6     ## TL;DR ? where we are
7
8     The **entire bucket-driven `train` ? infer` pipeline runs end-to-end on a CPU box**, with
9     the GPU **mocked**
10    (deterministic stub). Validated on a DigitalOcean droplet reading/writing **DO Spaces**,
11    driven remotely over
12    SSH. Everything is merged to `master` (PRs #246?#254); the suite is **936 green**.
13
14    **The ONE remaining piece for *real* (non-mock) training** is the **Linux-native WASB
15    `DetectorRunner` (M3.5)**
16    running on a **GPU box** ? it produces *real* shuttle trajectories instead of the
17    synthetic stub. Once it
18    exists, the **same `worker train` / `worker infer` commands run unchanged**; only the
19    trajectory source flips.
20
21    ## What's built (all merged to master)
22
23    | PR | What | Key files |
24    |---|---|---|
25    | #249 | Storage layer ? `local` + **S3 (AWS/DO Spaces/R2/MinIO)** + GCS, URI-driven;
26    gdrive stub | `backend/storage/` |
27    | #251, #254 | **Worker** `python -m backend.worker {infer|train}` ? pull from storage
28    ? pipeline ? push | `backend/worker/__main__.py` |
29    | #247 | CPU-mock detector seam (`RALLY_DETECTOR_IMPL=stub` /
30    `indexing.detector_impl=stub`) | `backend/pipeline/detectors/stub_runner.py` |
31    | #253 | GPU provisioning ? NVMe scratch + **native WASB setup** + port runbook |
32    `deploy/digitalocean/{bootstrap,mount_scratch,gpu_setup}.sh` |
33    | #252 | Cross-platform canonicalization (Linux/macOS, not just Windows) | (segmenters,
34    compiler, doctor.py) |
35    | #246 | The plan directory | `docs/DECOUPLED_COMPUTE/` |
36
37    **Verified e2e (CPU, no GPU, no Gemini key):** `worker train` pulled 7 labels from
38    Spaces ? `synth_trajectory_from_labels`
39    (label-consistent CPU mock) ? `training.gen0.harness` (shelled out ? `backend` must not
40    import `training`) ?
41    LOVO over 7 videos (F1 1.000 on the mock, gate=no-ship by design) ? `weights/` to
42    Spaces. `worker infer` pulled a
43    real video ? stub ? `outputs/<md5>/` to Spaces.
44
45    ## Resources & access (current)
46
47    - **CPU droplet:** `168.144.126.176` (region `blr1`, CPU-only, no NVMe). SSH: `ssh -i
48    ~/.ssh/do_rally root@?`
49    (key generated locally; pubkey added to DO). Code + venv at `~/rally`; Gemini key at
50    `~/root/.keys/GEMINI_API_KEY`.

```

```

35     - **DO Spaces:** bucket **`rally-e2e`**, region **`sgp1`**, endpoint
`https://sgp1.digitaloceanspaces.com`.
36     ? **The Spaces key was exposed in chat ? ROTATE it.** Creds live on the box at
`~/aws/credentials` (never in repo).
37     - **Corpus staged in Spaces:** `videos/` (8 .mp4) + `labels/` (7 `rallies.csv`) +
`weights/0.1.0-smoke/` + `outputs/`.
38     - **Drive corpus** (source): folder `1sHGL7iacnmM80ChT-p2cKy638gK504T_` (gdown-able by
file-id). 7 `[Done]` = video+labels;
39     **Video 1 `GX010135_proxy.mp4` is the held-out (non-Done) test**.
40     - **Helper scripts on the box:** `/root/do_e2e.sh` (reproduces the train+infer e2e),
`/root/do_spaces_config.json`
41     (`storage.s3` sgp1 + `skip_ai_handoff` + `detector_impl=stub` + permissive
validation), `/root/do_stage_{labels,videos}.sh`.
42
43     ## Reproduce the working e2e (today, CPU)
44
45     ```bash
46     ssh -i ~/.ssh/do_rally root@168.144.126.176 'bash ~/do_e2e.sh'
47     # train (7-video LOVO) + infer (real video) from Spaces ? artifacts back to Spaces.
48     ```
49
50     ## NEXT BUILD ? real trajectories (M3.5)
51
52     1. **GPU box** (owner): spin up a DO GPU Droplet **in `sgp1`** (co-located with the
bucket) ? `SETUP_NEW_MACHINE.md` §2
53     has the `doctl` command (verify slugs). Provide a DO API token if you want it
`doctl`-provisioned. Add the same SSH key.
54     2. **Port** (`SETUP_NEW_MACHINE.md` §9): git-archive ? scp ? `mount_scratch.sh` ?
`bootstrap.sh --gpu` ? `gpu_setup.sh`
55     (native WASB conda env) ? secrets (rotated Gemini key + Spaces creds).
56     3. **Build `NativeWasbRunner(DetectorRunner)`** in `backend/pipeline/detectors/` ?
sibling to `WasbRunner` but **without**
57     `WslCommandMixin`: run WASB in-process/native (no `wsl`/`conda
activate`/`to_wsl_mnt_path`), device `cuda/mps/cpu`,
58     weights via env (not `~/models`), emit the identical `Frame,Visibility,X,Y` CSV.
Select it via the existing
59     `detector_impl` seam (`trajectory_hybrid._resolve_runner`). Scout details: `06-RISKS-
AND-OPEN-QUESTIONS.md` (E1?E6 in the
60     merged workflow plan) + `docs/TRACKNET_WSL_SETUP.md`.
61     4. **Acceptance gate:** byte-parity ? same clip ? WSL CSV (Windows) == native CSV
(Linux) + identical candidate windows.
62     5. **Wire real extraction into `worker train`/`infer`:** today `worker train` uses
`synth_trajectory_from_labels`; add a
63     real-extract path that runs the detector per video ? trajectory CSV ? manifest. The
harness + push are unchanged.
64
65     ## Gotchas / open items
66
67     - **`backend` ? `training`** one-way wall ? the worker **shells out** to `python -m
training.gen0.harness`.
68     - **Gen-0 ship-gate F1@tIoU is provisional (#172)** ? advisory; do **not** hard-gate on
it.
69     - **WASB-C3** weight provenance is an open **legal** item before sharing any
trajectories/model externally.
70     - WASB (torch 1.11) and TrackNet (TF) must stay in **separate conda envs** (segfault if
co-installed).
71     - The box still has MinIO (`:9000/:9001`) + a GCS emulator (`:9023`) running from
earlier provider tests (harmless).
72     - Two-provider goal (GCS in a Bangalore-ish region, parallel hosting) and the
`/srv/khelsutra-site` web-hosting verify
73     + a real-Gemini-detection smoke are **queued, not done**.
74

```

I now have enough grounding. Let me quickly verify the storage base seam (the StorageBackend ABC) and the parity CI patterns to cite reuse precisely.

## 27. user (2026-06-20T09:00:22.541Z)

```
=== storage/base.py head ===
"""The ``StorageBackend`` ABC ? one interface for local + object-store + Drive.

Mirrors the codebase's ABC+Registry idiom (cf. ``SegmenterRegistry``). Concrete backends
live in sibling modules and lazy-import their cloud SDK so ``import backend.storage`` never
pulls ``boto3`` / ``google-cloud-storage`` / ``pydrive2``. See ``PLATFORM_ARCHITECTURE.md``
§3.1 and ``docs/DECOUPLED_COMPUTE/``.
"""

from __future__ import annotations

import os
from abc import ABC, abstractmethod
from typing import Iterator

from backend.storage.ref import StorageRef, StorageStat

class StorageBackend(ABC):
    """Backend-agnostic object storage. ``fetch_to_local`` + ``put`` are the two the
    worker dispatcher needs; the rest support listing/checkpointing/serving."""

    @abstractmethod
    def fetch_to_local(self, ref: StorageRef, dst: "str | os.PathLike[str] | None" = None,
                      *, overwrite: bool = False) -> str:
        """Make the object available as a local file; return its local path. For the
        ``local`` backend this is an identity passthrough (the object is already local)."""

    @abstractmethod
    def put(self, local_path: "str | os.PathLike[str]", ref: StorageRef, *,
           content_type: "str | None" = None) -> StorageRef:
        """Upload/copy a local file to ``ref``; return ``ref``."""

    @abstractmethod
    def exists(self, ref: StorageRef) -> bool: ...

    @abstractmethod
    def stat(self, ref: StorageRef) -> StorageStat: ...

    @abstractmethod
    def list(self, prefix: StorageRef) -> Iterator[StorageRef]:
        """Yield a ``StorageRef`` per object under ``prefix``."""

    @abstractmethod
    def delete(self, ref: StorageRef) -> None: ...

    def signed_url(self, ref: StorageRef, *, ttl: int = 3600, method: str = "GET") -> str:
        """A time-limited URL handing the bytes to a browser/LLM. Not every backend can
        sign (``local`` cannot; Drive only emulates) ? callers must treat
        ``NotImplementedError`` as expected and fall back to ``fetch_to_local``."""
        raise NotImplementedError(f"{type(self).__name__} cannot produce signed URLs")
=== storage/__init__.py (factory) ===
"""Pluggable storage layer: local (default) + S3/GCS first-class buckets + Drive (follow-up).

URI-driven (see ``ref.parse_uri``). The package import is cheap ? cloud SDKs are imported
lazily, only when a non-local URI is actually used (install the ``storage-s3`` / ``storage-gcs``
/ ``storage-gdrive`` extras). The default ``local`` backend needs no third-party deps and
preserves today's behaviour byte-for-byte.

Worker/dispatcher entry points speak URIs via the thin helpers ``fetch`` / ``put`` /
``list_uris`` / ``get_storage``; ``local://`` (and bare paths) make ``fetch`` an identity
```

```

passthrough. See ``docs/DECOUPLED_COMPUTE/``.
"""
from __future__ import annotations

import os
from typing import Any, Dict, Iterator, Optional

from backend.storage.base import StorageBackend
from backend.storage.ref import StorageRef, StorageStat, parse_uri

__all__ = [
    "StorageBackend", "StorageRef", "StorageStat", "parse_uri",
    "get_storage", "fetch", "put", "list_uris",
]

def _backend_class(name: str):
    """Lazy-resolve a backend class so cloud SDKs aren't imported until needed."""
    if name == "local":
        from backend.storage.local import LocalStorage
        return LocalStorage
    if name == "s3":
        from backend.storage.s3 import S3Storage
        return S3Storage
    if name == "gcs":
        from backend.storage.gcs import GCSStorage
        return GCSStorage
    if name == "gdrive":
        from backend.storage.gdrive import GDriveStorage
        return GDriveStorage
    raise ValueError(f"unknown storage backend: {name!r} (expected local|s3|gcs|gdrive)")

```

I have all the grounding I need. Let me look at the parity CI workflow briefly to cite the reuse pattern precisely.

```

1 # Live-host parity ? the site runs on TWO hosts (Cloudflare Worker + the DO droplet's
2 # Node/SQLite twin) that must serve byte-identical pages. The unit suite proves the code
3 # works on each host; it CANNOT see that the two LIVE URLs have drifted (e.g. a merge
4 # auto-deployed Cloudflare but the droplet wasn't re-synced from git). This runs the
5 # existing scripts/parity-check.sh against both live URLs on a schedule ? NOT on PRs, so
6 # it never blocks development; a failure surfaces real production drift to the owner.
7 name: live-parity
8
9 on:
10   schedule:
11     - cron: "17 6 * * *" # daily ~06:17 UTC
12   workflow_dispatch: {} # and on demand
13
14 # Avoid overlapping runs if a manual dispatch races the cron.
15 concurrency:
16   group: live-parity
17   cancel-in-progress: false
18
19 jobs:
20   parity:
21     runs-on: ubuntu-latest
22     steps:
23       - uses: actions/checkout@v4
24       - name: Compare the two live hosts byte-for-byte
25         env:
26           CF_URL: https://khelsutra.guru

```

```
27         VM_URL: http://168.144.126.176:8787
28     run: bash site/scripts/parity-check.sh
29
```

**30. assistant (2026-06-20T09:00:28.504Z)**

I have comprehensive grounding now. Let me compile the structured output.

**31. user (2026-06-20T09:01:43.258Z)**

Structured output provided successfully

**32. assistant (2026-06-20T09:01:46.963Z)**

Mapping complete. Structured output returned covering the four requested areas: the khelsutra UI surface (site/ public marketing vs web/ alpha SPA scaffold vs control-plane/ beta), the gated-app-vs-public-page boundary for the operator console, how per-rally selection is one screen of the broader console, the exact PROJECT\_DOC\_TEMPLATE structure to follow, concrete reuse opportunities, and path:line file refs.